Estimation of Job Vacancy Statistics Under Misclassification and Missing Data Using Capture-Recapture

Adam Struzik

Faculty of Mathematics and Computer Science, Adam Mickiewicz University in Poznań, Poland
Centre for Urban Statistics, Statistical Office in Poznań, Poland

Royal Statistical Society 2026 International Conference
Bournemouth, United Kingdom
10 September 2026

Table of contents

1 Introduction

2 Methodology
- Notation, data and assumptions
- Algorithm

3 Simulation study
- Data
- Methods and results
- Conclusions

Table of contents

1 Introduction

2 Methodology
- Notation, data and assumptions
- Algorithm

3 Simulation study
- Data
- Methods and results
- Conclusions

Overview

- A proof-of-concept approach for **estimating the number of job vacancies** based solely on online data and administrative registers.
- Uses **capture-recapture** and **latent class** analysis techniques.
- Corrects for the bias associated with **classification errors** in categorical covariates.
- Accommodates **missing and censored data**.

Vacancies

A **job vacancy** is defined by European Commission and Eurostat (2025) as a paid post that is newly created, unoccupied, or about to become vacant:

- for which the employer is taking active steps and is prepared to take further steps to find a suitable candidate from outside the enterprise concerned; and
- which the employer intends to fill either immediately or within a specific period of time.

Note. This definition may differ between countries.

Introduction

Measuring the size and characteristics of **the demand for labour**:

- is crucial for employers, businesses and economic policymakers;
- requires complementing employment statistics with **vacancy statistics** (Davis et al., 2013).

Vacancy data can also be used to study mismatch between job seekers and available jobs across occupations and sectors (Şahin et al., 2014).

Official methods for estimating the number of vacancies

Examples:

- Czechia: The estimation is based solely on data from the **administrative registers** of Labour Offices;
- Poland: The estimation is based on **survey sampling**; the Statistical Office in Bydgoszcz provides quarterly reports in the publication *The demand for labour* (DL; Statistics Poland, 2024).

For details regarding the European Union, see European Commission and Eurostat (JVS; 2025).

Disadvantages

- Czechia: Since 1 January 2012, it is no longer compulsory to report job vacancies to the Labour Office in Czechia.
- Poland: Beręsewicz et al. (2021) showed that DL underestimated the number of entities in Poland with at least one vacancy by about 10–15%; the reasons are increasing **non-response** and **under-reporting** rates.

Online job advertisements (OJAs) and administrative data as an alternative

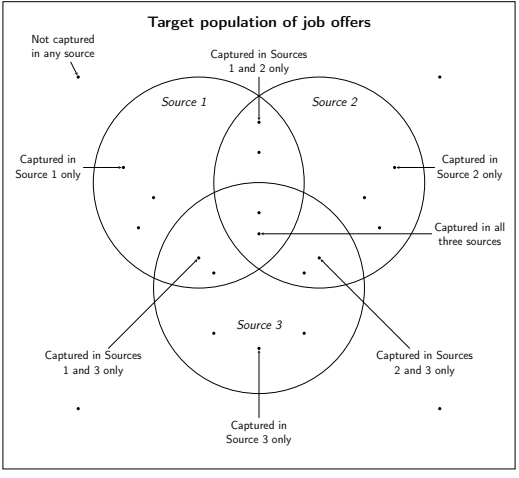
Motivation:

- **Vast amounts of online job advertisements** contain valuable information.

Challenges:

- OJAs represent **only a part of the target population** (Napierała et al., 2022).
- OJAs often **lack information about occupation codes** in accordance with the International Standard Classification of Occupations (ISCO).
- OJAs **may not include the exact number of advertised vacancies**. One observes statements such as 'we will hire many people'.

Solution: Capture-recapture (CRC) methods



Polish labour market context

- The Polish public register-based job portal ePraca always reports:
  - occupation codes in accordance with ISCO,
  - exact numbers of free job positions.
- In the proposed method, Source 1 and the assumptions imposed on it correspond to ePraca.
- For Sources 2 and 3, occupation codes, if missing, can be predicted by a machine learning classifier.

Table of contents

1 Introduction

2 Methodology
- Notation, data and assumptions
- Algorithm

3 Simulation study
- Data
- Methods and results
- Conclusions

Notation and data

We consider a finite population of job offers $U = \{1, 2, \ldots, N\}$, where N is unknown, and three job portals. For $i \in U$, we define:

- $\boldsymbol{Y}_i = (Y_{i1}, Y_{i2}, Y_{i3}) \in \{0, 1\}^3$ – the capture history,
- $G_i \in \{1, 2, \ldots, C\}$ – the true (often unknown) occupation code,
- $X_i \in \{1, 2, \ldots, K\}$ – the latent class (membership in a hidden population stratum),
- $V_i \in \{1, 2, \ldots\}$ – the number of positions (vacancies) in a job offer.

We want to estimate **the total number of vacancies**, i.e.

$$T = \sum_{i=1}^{N} V_i.$$

Notation and data

- Let S denote the set of observed job offers, i.e.

$$S \coloneqq \{i \in U : \boldsymbol{Y}_i \neq (0,0,0)\}.$$

- Let

$$N_{\boldsymbol{y},g,x} \coloneqq \#\{i \in U : \boldsymbol{Y}_i = \boldsymbol{y}, G_i = g, X_i = x\}.$$

- Let $\tilde{G}_i$ and $\tilde{V}_i$ denote the observed (potentially predicted) occupation code and the observed number of vacancies, respectively.
- Source 1 $\sim$ ePraca implies that if $Y_{i1} = 1$, then $\tilde{G}_i = G_i$ and $\tilde{V}_i = V_i$.
- Otherwise, we might have $\tilde{G}_i \neq G_i$ (**misclassification**) and the information on V_i might be **missing or censored** at 2+ (e.g. 'we will hire many people').
- Let S_{val} be an independent validation sample (disjoint from S), including both G_i and $\tilde{G}_i$.

Fundamental assumptions

- There exists **a constant confusion matrix $\mathbf{\Pi} \in [0,1]^{C \times C}$** such that for the units $i \in S$, we have

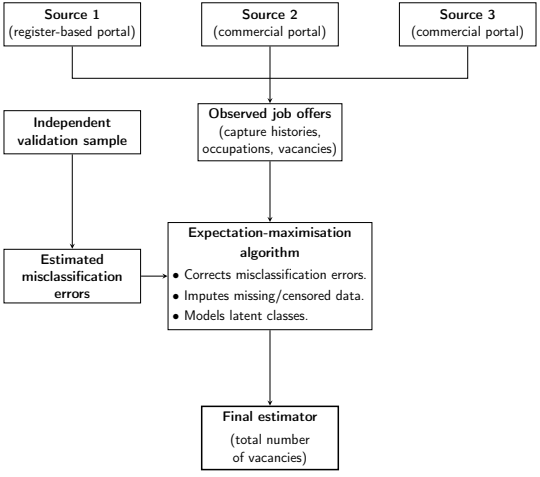
$$\mathbb{P}(\tilde{G}_i = h \mid G_i = g, Y_{i1} = 0) = \Pi_{gh}, \quad g, h \in \{1, 2, \ldots, C\}.$$

- For each combination $(\boldsymbol{y}, g, x)$, there exists an expected cell frequency $m_{\boldsymbol{y},g,x} > 0$, and the cell counts are independent and follow a Poisson distribution, i.e. $N_{\boldsymbol{y},g,x} \sim \text{Poisson}(m_{\boldsymbol{y},g,x})$, leading to the following **log-linear model** (cf. Baffour et al., 2021):

$$\log m_{\boldsymbol{y},g,x} = \lambda + \lambda_g^G + \lambda_x^X + \sum_{s=1}^{3} \lambda_{y_s}^{(s)} + \sum_{s=1}^{3} \lambda_{y_s,x}^{(sX)} + \lambda_{g,x}^{(GX)} + \lambda_{y_1,y_2}^{(12)} + \lambda_{y_1,y_3}^{(13)}.$$

- Given $G_i = g$, V_i follows a zero-truncated negative binomial distribution with parameters μ_g and σ (cf. Brännäs, 1992).

Flowchart of the method



Estimation of the confusion matrix and likelihood function

- Using the validation sample S_{val}, we calculate $\hat{\mathbf{\Pi}}$ as an estimate of $\mathbf{\Pi}$.
- Let $f_g(\cdot; \mu_g, \sigma)$ denote the probability mass function of a zero-truncated negative binomial distribution with parameters μ_g and σ. For $i \in S$, we define

$$L_V(i \mid g; \mu_g, \sigma) = \begin{cases} f_g(\tilde{V}_i; \mu_g, \sigma), & \text{if } \tilde{V}_i = V_i, \\ \mathbb{P}_{\mu_g,\sigma}(V_i \geq 2 \mid G_i = g), & \text{if } \tilde{V}_i = 2+, \\ 1, & \text{if } \tilde{V}_i \text{ is missing.} \end{cases}$$

Estimation of the confusion matrix and likelihood function

- Let $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$ denote the parameters associated with the log-linear model and with modelling the number of vacancies, respectively.
- Conditional on $\hat{\boldsymbol{\Pi}}$, the objective function used in the proposed EM algorithm consists of **two components**, given by:

$$\ell_c^{(Y)}(\boldsymbol{\theta}) = \sum_{\boldsymbol{y},g,x} [N_{\boldsymbol{y},g,x} \log m_{\boldsymbol{y},g,x}(\boldsymbol{\theta}) - m_{\boldsymbol{y},g,x}(\boldsymbol{\theta})] + \text{const}$$

and

$$\ell^{(V)}(\boldsymbol{\phi}) = \sum_{i \in S} \log L_V(i \mid G_i; \boldsymbol{\phi}).$$

Single iteration

The t-th iteration is as follows:

- **E-step**: Calculate

$$\tau_{i,g,x}^{(t)} = \mathbb{P}(G_i = g, X_i = x \mid \boldsymbol{Y}_i, \tilde{G}_i, \tilde{V}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}})$$

  and

$$N_{\boldsymbol{y},g,x}^{(t)} = \mathbb{E}[N_{\boldsymbol{y},g,x} \mid \{(\boldsymbol{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}].$$

- **M-step**: Calculate

$$(\boldsymbol{\theta}^{(t+1)}, \boldsymbol{\phi}^{(t+1)}) = \left(\arg\max_{\boldsymbol{\theta}} \mathbb{E}\left[\ell_c^{(Y)}(\boldsymbol{\theta}) \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}\right], \arg\max_{\boldsymbol{\phi}} \mathbb{E}\left[\ell^{(V)}(\boldsymbol{\phi}) \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}\right] \right).$$

Full algorithm

1. **Input**: $\{(\boldsymbol{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}$, $\hat{\boldsymbol{\Pi}}$.
2. **Initialisation**: Initialise $N^{(0)}_{(0,0,0),g,x}$ and $\tau^{(0)}_{i,g,x}$, and calculate $N^{(0)}_{\boldsymbol{y},g,x}$ for $\boldsymbol{y} \neq (0,0,0)$.
3. For $t = 0, 1, \ldots$, repeat until convergence:
   1. **M-step**: Obtain $\boldsymbol{\theta}^{(t+1)}$ and $\boldsymbol{\phi}^{(t+1)}$ by solving the maximisation tasks.
   2. **E-step**: Calculate $\tau^{(t+1)}_{i,g,x}$ and update $N^{(t+1)}_{\boldsymbol{y},g,x}$.
4. **Output**: The final estimated parameters $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$, along with $\hat{\tau}_{i,g,x}$ and $\hat{N}_{\boldsymbol{y},g,x}$.

Estimator of the total number of vacancies

For $i \in S$, we define

$$V'_{i,g} = \begin{cases} \tilde{V}_i, & \text{if } \tilde{V}_i = V_i, \\ \mathbb{E}_{\hat{\boldsymbol{\phi}}}[V \mid V \geq 2, G = g], & \text{if } \tilde{V}_i = 2+, \\ \mathbb{E}_{\hat{\boldsymbol{\phi}}}[V \mid G = g], & \text{if } \tilde{V}_i \text{ is missing.} \end{cases}$$

We estimate the total number of vacancies as follows:

$$\hat{T} = \sum_{i \in S} \sum_{g=1}^{C} \sum_{x=1}^{K} \hat{\tau}_{i,g,x} V'_{i,g} + \sum_{g=1}^{C} \sum_{x=1}^{K} \hat{N}_{(0,0,0),g,x} \mathbb{E}_{\hat{\boldsymbol{\phi}}}[V \mid G = g].$$

Table of contents

1 Introduction

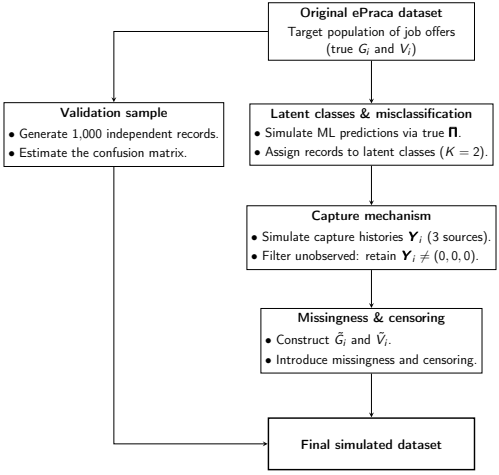
2 Methodology
- Notation, data and assumptions
- Algorithm

3 Simulation study
- Data
- Methods and results
- Conclusions

Simulation study

- The proposed approach has been implemented in R (R Core Team, 2026).
- We use all job offers available on ePraca in January 2026 (except for one related to the military).
- We consider the first digit of the occupation code.
- The resulting dataset has 10,524 records, and the total number of reported vacancies is 21,532.
- We adopt the confusion matrix reported by Beręsewicz et al. (2026, Table 11, Large model) as $\mathbf{\Pi}$.

Data generation mechanism



Compared methods

- A naïve two-step approach (*M1*): The method by Baffour et al. (2021) and a zero-truncated negative binomial regression model applied to the observed data.
- The proposed approach, but with the use of the true confusion matrix $\mathbf{\Pi}$ (*M2*).
- The proposed approach with the use of the estimated confusion matrix (*M3*).

Results of the simulation study

Average estimate of the number of vacancies, its mean bias, relative mean bias and root mean square error over 100 simulations.

Method	$\hat{T}$	MB	MRB	RMSE
M1	20,937.32	-594.68	-2.76%	1,234.53
M2	21,597.01	65.01	0.30%	1,286.64
M3	21,648.32	116.32	0.54%	1,370.56

Note. $\hat{T}$: the average estimate of the number of vacancies over 100 simulations; MB: mean bias; MRB: mean relative bias; RMSE: root mean square error. The code and data are available on GitHub.

Conclusions

- The proposed method **substantially decreases the bias** present in the naïve approach.
- Relatively small discrepancies between models *M2* and *M3* confirm the effectiveness of using an external validation sample.
- The increase in RMSE for the proposed method reflects a standard bias-variance trade-off.

Possible extensions and future directions

Possible extensions:

- Modifying the interaction terms between data sources.
- Adding **more categorical covariates** (e.g. related to employers).
- Applying alternative count distributions for the number of vacancies.
- Using different censoring thresholds (e.g. 'we are looking for at least 3 people').

Future directions:

- Addressing linkage errors.
- Incorporating sampling uncertainty associated with the estimation of the confusion matrix.
- Incorporating skills classification.

Acknowledgements

The research has been supported by the National Science Centre in Poland, under the project *OJALAB: Online job advertisements to study skill demand and job search patterns*, grant number 2024/53/B/HS4/01580.

Thank you!

Adam Struzik

Faculty of Mathematics and Computer Science, Adam Mickiewicz University in Poznań, Poland
Centre for Urban Statistics, Statistical Office in Poznań, Poland

adastr5@st.amu.edu.pl

References I

Baffour, B., Brown, J. J., and Smith, P. W. (2021). Latent class analysis for estimating an unknown population size – with application to censuses. *Journal of Official Statistics*, 37(3):673–697.

Beręsewicz, M., Cherniaiev, H., and Pater, R. (2021). Estimating the number of entities with vacancies using administrative and online data.

Beręsewicz, M., Wydmuch, M., Cherniaiev, H., and Pater, R. (2026). Multilingual hierarchical classification of job advertisements for job vacancy statistics. *Journal of Official Statistics*, 42(1):23–61.

Brännäs, K. (1992). Limited dependent Poisson regression. *Journal of the Royal Statistical Society. Series D (The Statistician)*, 41(4):413–423.

Davis, S. J., Faberman, R. J., and Haltiwanger, J. C. (2013). The establishment-level behavior of vacancies and hiring. *The Quarterly Journal of Economics*, 128(2):581–622.

References II

European Commission and Eurostat (2025). Job vacancy statistics (jvs) – reference metadata. `https://ec.europa.eu/eurostat/cache/metadata/en/jvs_esms.htm`. Accessed: 2026-03-28.

Napierała, J., Kvetan, V., and Branka, J. (2022). *Assessing the representativeness of online job advertisements*. Publications Office of the European Union.

R Core Team (2026). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Statistics Poland (2024). Methodological report. The demand for labour. Technical report, Statistics Poland.

Şahin, A., Song, J., Topa, G., and Violante, G. L. (2014). Mismatch unemployment. *American Economic Review*, 104(11):3529–64.